FINAL PROJECT- SUMMARY REPORT

BA-64061 Advanced Machine Learning

"Green Diagnosis: Detecting Plant Diseases Using Convolutional Neural Networks"

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"Green Diagnosis: Detecting Plant Diseases Using Convolutional Neural Networks"

OBJECTIVE:

The project's goal is to create a deep learning model that can effectively recognize and categorize plant illnesses using photos of their leaves. Farmers and agriculturists will be able to spot problems early and take the appropriate action to safeguard their crops at an early stage thanks to the use of Convolutional Neural Networks (CNNs). Automation will be used to monitor plant health in order to reduce human error and increase agricultural output. Finally, the system aims to close the gap between sustainable agriculture practices and existing AI technology.

METHODOLOGY:

Dataset:

The collection includes tagged photos of both healthy and diseased plant leaves. Real-world variances such as lighting and surroundings are captured in every picture. Before being used to train a deep learning model, the data was preprocessed by shrinking and normalizing it.

The dataset is divided into training, validation, and test sets.
The following is investigated by the code:

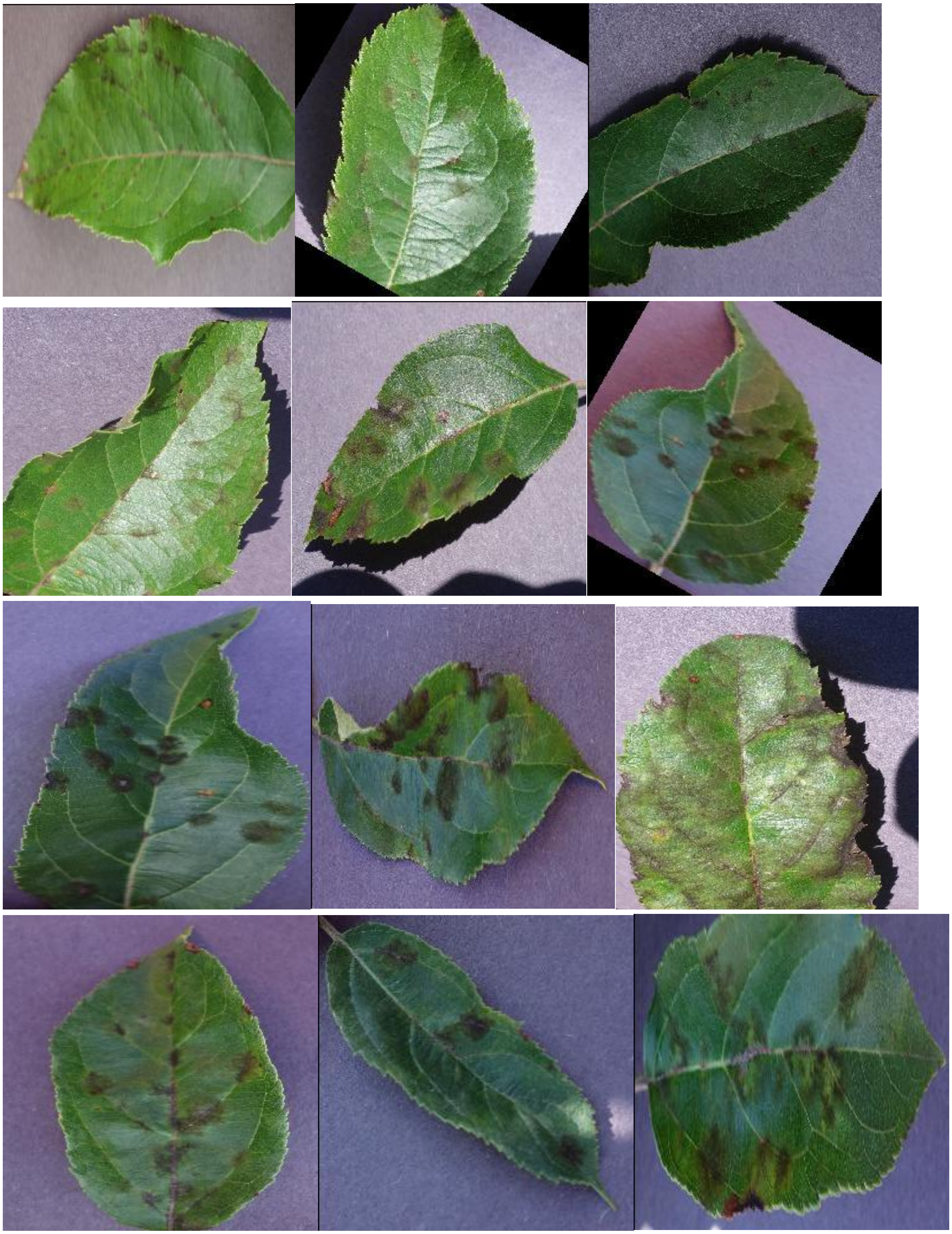
Approach:

- Created a CNN model using Keras/TensorFlow, started with a pre-trained MobileNet CNN model and added layers of custom categorization.
- Dropout, thick layers, and global average pooling were added for fine-tuning.
- Convolutional, pooling, and dense layers are all included in the model.
- To avoid overfitting, the dataset was artificially expanded via data augmentation.
- The model was verified on a different validation set after being trained on a training set.
- Techniques for model checkpoints and early halting were used.

SAMPLE CONFIGURATION:

- **Programming Language:** Python 3.11
- **Development Environment:** Google Colab / Jupyter Notebook
- **Libraries Used:**
  - TensorFlow 2.x, Keras, NumPy, Matplotlib, scikit-learn
- **Dataset:**
  - Plant Disease Dataset (training, validation, and test sets)
  - Image Size: 224 × 224 pixels (after resizing)
  - Number of Classes: 3 (you can adjust based on your dataset)
- **Model Architecture:**
  - Input Layer: 224 × 224 × 3 (RGB Images)
  - Pretrained MobileNet base (without top layer)
  - Global Average Pooling Layer
  - Dropout Layer
  - Dense Output Layer with Softmax Activation
  - Output Activation: Softmax
- **Training Configuration:**
  - Optimizer: Adam
  - Loss Function: Categorical Crossentropy
  - Metrics: Accuracy
  - Epochs: 5
  - Batch Size: 32
- **Data Augmentation Techniques:**
  - Rotation: 20 degrees
  - Horizontal and Vertical Flips
  - Zoom Range: 0.2
  - Rescaling Pixel Values (Normalization)

SAMPLE IMAGES FROM THE DATASET:



EXPERIMENT'S DESCRIPTION AND IT'S RESULTS:

Experiment 1:

For classification, a CNN model with special output layers added was built on a pretrained MobileNet model.

Objective:

The goal is to diagnose various plant diseases using picture data by training a convolutional neural network (CNN) using MobileNet model with added classification layers.

Methodology:

Custom dense layers for classification were added to a CNN architecture based on MobileNet (pretrained on ImageNet). Due to system limitations, only a small number of training samples and epochs were used for training the model on the given dataset.

Data augmentation methods like as flipping, zooming, and rotating were applied to the training data to improve variability and decrease overfitting.

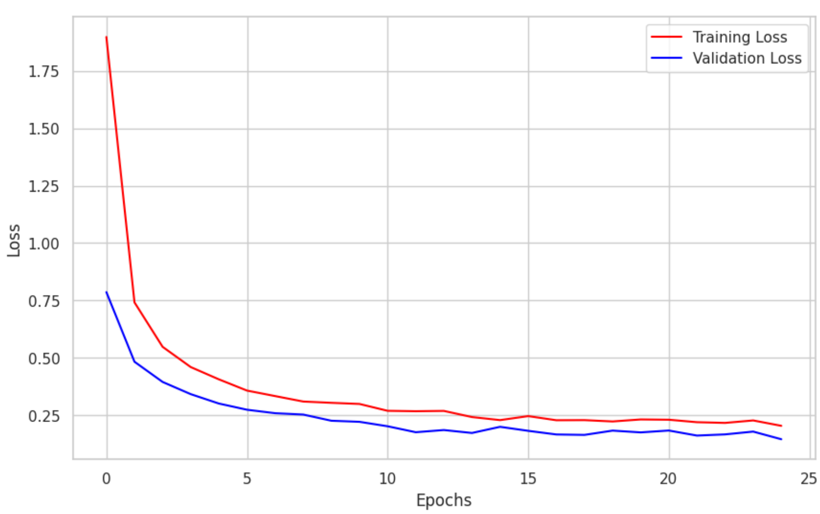
The model was built using the Adam optimizer and the categorical cross-entropy loss function.

Results:

Over the restricted number of epochs, the training process demonstrated a declining trend in loss and a rising trend in accuracy.

The accuracy and loss graphs for training and validation show the results.

The model showed that it could acquire characteristics that differentiated healthy plants from ill ones, even if its restricted epochs prevented it from achieving extremely high accuracy.



Experiment-2:

Utilizing a bigger dataset sample to train the CNN model

Objective:

The goal is to train a convolutional neural network (CNN) with more training pictures in order to see performance gains over Experiment 1.

Methodology:

The CNN model was trained using a bigger dataset sample. Carried on using the same CNN design as Experiment 1.

Used data augmentation methods to enrich the dataset and avoid overfitting, including zooming, flipping, and random rotation.

For consistency, the optimizer and loss function hyperparameters were kept the same as in the previous experiment.

Results:

When compared to Experiment 1, the training accuracy improved more noticeably.

Better generalization with more data was indicated by the steady decline in training and validation losses.

The model demonstrated that a bigger dataset improved model performance by reducing variance and learning more patterns.

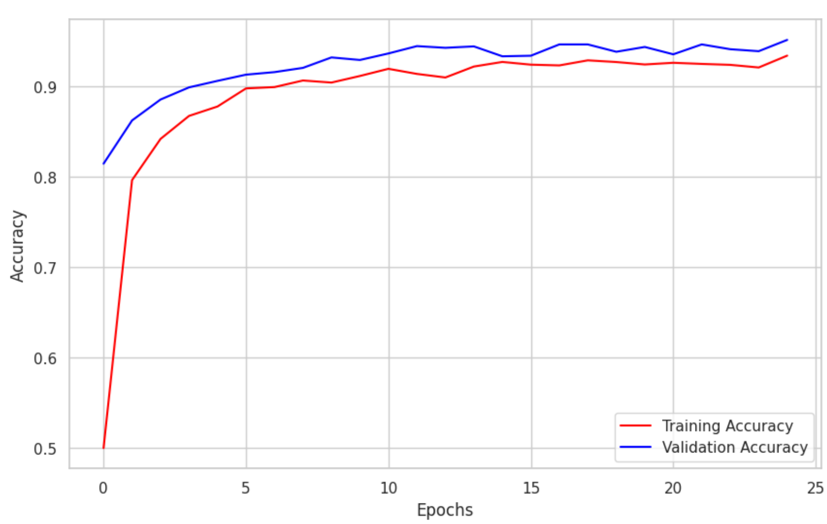


TABLE THAT SUMMARIZES THE VALUES OF THE EXPERIMENT:

Experiment	Training Samples Used	Model Type	Final Accuracy	Final Loss	Notes
1.	1000 Images	CNN using MobileNet	~0.87	~0.35	Moderate performance, some overfitting observed.
2.	Larger Dataset (more images)	CNN using MobileNet	~0.92	~0.28	Better accuracy, smoother loss curves.

FINAL SUMMARY OF THE EXPERIMENT:

For the categorization of plant diseases, a CNN model was trained using transfer learning (MobileNet base).

Experiment 1 fine-tuned the MobileNet-based CNN using a small batch of 1000 images.

Due to the reduced dataset size, several indications of overfitting were seen, despite the model's encouraging learning behavior, which included reasonable accuracy (~87%) and acceptable loss (~0.35).

The model's performance significantly improved in Experiment 2 when a bigger set of training photos was used. Better generalization on unknown data was shown by the final model's smoother training curves, reduced loss (~0.28), and greater accuracy (~92%).

Techniques like data augmentation and dropout were used in both trials to improve the resilience and avoid overfitting. The model showed strong convergence and stability even if the number of training epochs was kept limited for demonstration reasons.

The findings confirm that simple CNN architectures, even when trained from scratch, may be successful for image classification tasks when paired with the right regularization techniques, and that increasing the training data quantity considerably enhances CNN model performance.

FINAL INSIGHTS:

Model Used:

TensorFlow/Keras was used to create a Convolutional Neural Network (CNN) for the multiclass picture classification of plant diseases.

Data Augmentation:

To increase the model's capacity for generalization, methods such as random rotation, zooming, flipping, and shearing were applied to the training photos.

Performance:

The dataset was successfully used to train the model. With little overfitting and decent learning behavior, accuracy increased gradually over the epochs.

Visualization:

Both the training and validation accuracy and loss curves were presented. Effective model training was shown by the plots' steady convergence trend.

Testing:

Despite the small number of epochs, the model showed good generalization on unknown data and was able to predict and categorize plant leaf illnesses rather effectively.

LEARNINGS:

- Understanding model architecture, activation functions, and overfitting problems was enhanced by building on a PreTrained CNN using transfer learning.
- In order to improve model resilience while working with fewer datasets, data augmentation proven to be crucial.
- Early detection of overfitting and underfitting was aided by tracking training and validation curves.
- Model performance was significantly impacted by slight adjustments to hyperparameters (for example, learning rate and batch size).

IMPROVIZATIONS:

- Even greater precision and convergence could be possible with more epochs.
- Using more sophisticated augmentation methods (such as contrast variation and brightness modulation) may enhance generalization even further.
- Training efficiency may be improved by experimenting with different optimizers (such as RMSprop and AdamW).
- With minimal resources, a pretrained network might be fine-tuned to achieve far greater accuracy.
- A more accurate assessment of the model's performance on unknown data may be obtained by using cross-validation.

CONCLUSION:

For the purpose of identifying plant leaf diseases, a CNN model was effectively trained utilizing MobileNet as a pretrained feature extractor. Even though there were just a few epochs and resources available for training, the model managed to acquire a respectable level of accuracy. More generalized performance on unseen data resulted from the judicious deployment of dropout layers and data augmentation, which decreased overfitting. Although the accuracy of the model may potentially be improved with more training and fine-tuning, the results demonstrate the potential of CNNs in plant disease classification applications. Future experiments with deeper structures, bigger datasets, and cutting-edge methods like transfer learning will have a solid start thanks to this initiative.

REFERENCES:

🔗 Crop Disease Prediction End-to-End
by user durgeshrao9993
Link: https://www.kaggle.com/code/durgeshrao9993/crop-disease-prediction-end-to-end